

Ultrasound-Induced Prenatal Stress: New Possibilities for Modeling Mental Disorders

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Keywords

Prenatal stress · Ultrasound exposure · Mental disorders · Animal models · Neurotransmitters

Abstract

The development of animal models of mental disorders is an important task since such models are useful for studying the neurobiological mechanisms of psychopathologies and for trial of new therapeutic drugs. One way to model pathologies of the nervous system is to impair fetal neurodevelopment through stress of the pregnant future mother, or prenatal stress (PS). The use of variable frequency ultrasound (US) in rodents is a promising method of imitating psychological stress, to which women in modern society are most often subjected. The aim of our study was to investigate the effect of PS induced by exposure to variable frequency ultrasound (US PS) throughout the gestational period on the adult rat offspring, namely, to identify features of behavioral alterations and neurochemical brain parameters that can be

associated with certain mental disorders in humans, to determine the possibility of creating a new model of psychopathology. Our study included a study of some behavioral characteristics of male and female rats in the elevated plus maze, open-field test, object recognition test, social interaction test, sucrose preference test, latent inhibition test, Morris water maze, forced swimming test, acoustic startle reflex, and prepulse inhibition tests. We also determined the activity of the serotonergic, dopaminergic, and noradrenergic neurotransmitter systems in the hippocampus and frontal cortex by HPLC-ED. Concentration of norepinephrine, dopamine, DOPAC, serotonin, and HIAA, as well as DOPAC/dopamine and HIAA/serotonin ratios were determined. A correlation analysis of behavioral and neurochemical parameters in male and female rats was performed based on the data obtained. The results of the study showed that US PS altered the behavioral phenotype of the rat offspring. US PS increased the level of anxious behavior, impaired orientation-research behavior, increased grooming activity, decreased the desire for social contacts, shifted behavioral

reactions from social interaction to interaction with inanimate objects, impaired latent inhibition, and decreased the startle reflex. US PS activated the serotonergic, dopaminergic, and noradrenergic neurotransmitter systems of the rat frontal cortex and hippocampus. A correlation between neurochemical and behavioral parameters was revealed. Our study showed that US PS leads to a certain dysfunction on behavioral and neurochemical levels in rats that is most closely associated with symptoms of schizophrenia or autism. We hypothesize that this could potentially be an indicator of face validity for a model of psychopathology based on neurodevelopmental impairment.

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Introduction

“Stress” is “a non-specific response of the organism to any demand for change” as defined by Hans Selye [1, 2]. Selye identified several stages of the organism’s stress response, the last of which “exhaustion” is potentially dangerous [3]. The consequences of stress can include both negative and positive aspects [4, 5]. An optimal level of stress is necessary to create adaptation to negative influences through hormesis, which involves a positive response to a low dose of stress [4, 6]. However, disruption of normal physiological function occurs when all stimuli outweigh the ability to cope with the effects of stress [4, 5, 7]. Stress can be caused by various factors including environmental factors (light, noise, irritating colors, uncomfortable temperature, lack of access to food and water, crowding, negative social experiences) [7], in which case these factors are considered negative environmental stressors. Stress exhaustion can lead to various behavioral disorders, among others, for example, increased population density is a strong stressor that can induce a stressful physiological response in mammals [8], and even lead to aberrant behaviors such as cannibalism [9], increased aggression [10], addictions in humans [11].

A separate area is the study of stress in early life, including the prenatal period. Prenatal period of organism development is an incredibly important stage because it is very sensitive to the various positive (enriched environment) and negative stimuli (maternal deprivation, malnutrition, social isolation) [12–15]. These factors together with genetic predisposition produce a complex effect on the structural and functional organization of the entire organism, particular of the brain, which finally determines the behavioral phenotype of the organism [13–16]. Nervous system development

can disrupt after external deleterious environment exposure and induce psychopathology manifestation [17]. Stress experienced by the mother during pregnancy is described as prenatal stress (PS). It is one of the main negative environmental factors, which can indirectly affect the offspring [13, 18].

A large number of studies describe the effects of PS on the offspring in humans and other animals [17]. Stress induction in pregnant female animals leads to the development of a pathological behavioral phenotype in the offspring. Stressful life events during pregnancy, as well as symptoms of maternal anxiety and depression, increase the risk of a number of emotional, behavioral, and cognitive problems in the child [19]. Experienced PS is associated with mental disorders such as schizophrenia [20], depression, anxiety [21], autism [22, 23], and attention deficit hyperactivity disorder [22]. With this background, studying the effects of PS is an important task [24].

It is also important to note that the study of PS via the animal models can be advantageous as a tool for modeling of psychopathologies. Given that neurobiological studies of human psychiatric pathologies are limited, the using of animal models for the investigation of brain impairments is practicable [25]. As PS disrupts the neurodevelopment, the usage of PS has the potential to imitate major etiological aspects of certain mental disorders [26]. Therefore, animal models based on PS exposure are applicable for the research of the neurobiological mechanisms of psychopathologies for the evaluation of the new therapeutic drugs.

PS can alter not only behavioral traits but also related biochemical parameters, which reflect the functioning of neurotransmitter systems of the brain. Neurotransmitters play the critical role in the process of information transmission between the neurons in the nervous system [27]. The impairments of neurotransmitter systems can induce behavioral alterations, leading to the development of psychiatric pathologies [28–30]. Serotonin, dopamine, and norepinephrine are the most studied and functionally important neurotransmitters [31]. Serotonin is one of the main monoamine neurotransmitter in the nervous system. Serotonergic neurons modulate a wide range of functions, including mood, perception, aggression, anxiety, cognition, attention, circadian rhythms, and embryonic development. Disruption of the serotonergic system is critical for the manifestation of several mental disorders, including anxiety, depression, and schizophrenia [32]. Dopamine and norepinephrine represent catecholamine neurotransmitters. The dopaminergic system is important for the neuromodulation of

motivation, reward, cognitive functions, and reproductive behavior [33]. Altered catecholamine transmission play the significant role in the etiology of psychiatric disorders [34]. In particular, the dopamine hypothesis is the longest-standing pathoetiological theory of schizophrenia [28, 35]. Norepinephrine is produced predominantly by noradrenergic neurons in the locus coeruleus and is released in almost all brain regions. It is involved in many processes, including learning, memory and attention, mood, and stress response. It is also considered as an important regulator of the brain development [36] and associated with cognitive impairment and schizophrenia [37].

Our study is focused on the new approach to rodents stressing. Basically, stress exposure can be of a different nature, and can often be divided into physical and psychological stress. In the modern society, pregnant women experience psychological stress more often than physical stress. Consequently, the using of the animal models based on psychological stress exposure could be preferable [38]. Various exposure modalities are widely used to reproduce PS in animals for scientific research, such as restraint stress or chronic unpredictable mild stress [26]. However, these approaches do not focus on the main stressor that affects humans in the modern society: the informational stress. The using of variable frequency ultrasound (US) exposure in rats seems promising for this purpose as adult rodents vocalize 22–25 kHz US signals during dangerous situations, and pups vocalize 40 kHz signals when isolated from their dams [39]. Earlier, we demonstrated that unavoidable US exposure induces stress condition in rodents, and when chronically exposed for 21 days leads to the development of depressive-like state [40–44]. It should be noted that our model of US stress differs from another model based on sound exposure, the noise stress model [45]. This type of stress is based only on noise exposure to “white noise,” which is often presented at high intensity. However, our model of US stress is based on the alternation of different US frequencies of moderate loudness (50 dB), which create primarily an information load for rodents and thus the animals experience information stress.

Therefore, we hypothesized that chronic exposure of pregnant females to US for the entire gestation period could eventually affect the offspring. Indeed, our studies have demonstrated that PS induced by the US exposure (US PS) alters the behavior of both adult and juvenile rats [46, 47]. In the current work, we aimed to study more thoroughly the behavioral phenotype of adult rats after US PS, to determine the effect of US PS on the brain neurotransmitter systems, and to evaluate their associ-

ations. We suggest that some identified traits of the behavioral and biochemical phenotype of rats after US PS can be compared to the symptoms of certain human psychiatric disorders (schizophrenia spectrum disorder or autism). This approach can allow us to determine the possibility of modeling of psychopathologies on the basis of US PS exposure and to demonstrate the face validity of the prospective translational model.

Materials and Methods

Experimental Animals

We used the rats from the Nursery for Laboratory Animals (Pushchino, RAS, Moscow region) in the experiment. The experimental part of the work was performed on 20 female Wistar rats from which the offspring were obtained ($n = 165$). Housing conditions and all experimental procedures were conducted in accordance with the Directive 2010/63/EU of September 22, 2010 and approved by the Local Ethical Committee of V.P. Serbsky National Medical Research Center for Psychiatry and Narcology, approval number 1 from February 10, 2022. All animals were kept in the laboratory vivarium with constant room temperature 22°C, light regime 12/12 h, and free access to food and water (*ad libitum*). Before the start of the experiments, the animals underwent the period of adaptation for 7 days.

Design of the Experiment

The scheme of the experiment is shown in Figure 1. Three months old female rats were randomly paired with males at the rate of two females per male. Vaginal swabs were sampled daily from the females to determine the moment of fertilization. The day the sperm was detected in the swab was considered as the 0 day of pregnancy. Female rats from each male were randomly divided into two groups after the fertilization and placed in individual cages (53 cm × 35 cm × 19 cm). The females from which the experimental genotype (PS offspring) were subsequently obtained ($n = 13$) were placed under the experimental US radiation immediately after the fertilization for the entire period of gestation. Females from which the control offspring were obtained ($n = 7$) were kept in individual cages under the standard conditions without experimental exposure for the entire gestation period. All females after parturition were kept in individual cages with their pups under the standard conditions without experimental exposure. The day of the birth was considered as the 1st postnatal day (PND) for the newborn rats. Pups were weaned from the dams at the age of 21 PND, separated by sex, and divided into the groups of not more than 10 animals in 1 cage until the maturation (age 2 months). Therefore, four groups of adult animals were obtained: the control males, the control females, PS males, and PS females.

The resulting adult control and PS offspring at the age 60 PND were divided randomly into the two groups to minimize the negative effects of a large number of behavioral tests. The following behavioral tests were performed with the first group of offspring (12 control males, 14 control females, 13 PS males, 14 PS females) in the following sequence: elevated plus maze test (EPM) (70 PND), open-field test (74 PND), object recognition test (75 PND), sucrose preference test (80 PND), social interaction test (83 PND),

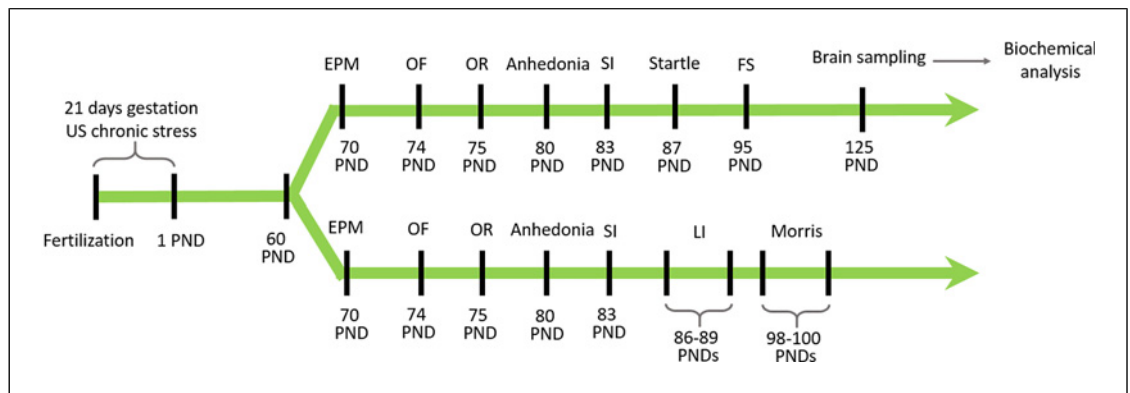


Fig. 1. Scheme of the experiment. Anhedonia – sucrose preference test; OF, open-field test; EPM, elevated plus maze; OR, object recognition test; SI, social interaction test; Startle, startle reflex test; FS, forced swimming test; LI, latent inhibition test; Morris, Morris water maze; PND, postnatal day; US, variable frequency ultrasound.

startle reflex test (87 PND), and forced swimming test (95 PND). The animals were not subjected to any manipulations for 1 month after the last test. After that, the animals from this group were sacrificed at the age of 125 PND for the brain tissue sampling.

The following behavioral tests were performed with the second group of animals (23 control males, 23 control females, 31 PS males, 35 PS females) in the following sequence: EPM (70 PND), open field test (74 PND), object recognition test (75 PND), sucrose preference test (80 PND), social interaction test (83 PND), latent inhibition test (86–89 PND), and Morris water maze test (98–100 PND).

Experimental Exposure to Variable Frequency US

PS in pregnant female rats was induced by the exposure to the US of variable frequency. The US was applied permanently during the whole period of gestation (21 days) using the US generator with the following technical specifications: power supply – 220 V through the device adapter, power consumption – 0.3 W. Sound pressure level was $50 \text{ dB} \pm 5 \text{ dB}$ (fluctuation $\pm 10\%$). The range of US frequency was alternated every 10 min between the following frequencies: short frequencies (20–25 kHz), medium frequencies (25–40 kHz), high frequencies (40–45 kHz). The total duration of radiation at short and medium frequencies was 35%, at high frequencies – 30%. The generator was placed 1 m above the cages with the animals, and the location of the cages was changed daily [41, 43, 44, 46].

Behavioral Tests

Behavioral testing was conducted between 10:00 a.m. and 2:00 p.m. Animals were transferred to the room where behavioral tests were performed at least 30 min before the testing. The studied parameters were registered under the standard conditions at an air temperature of 22°C and under the white light (100 lx) or red light, depending on the test conditions. Animal's behavior was registered via the digital video camera and analyzed using the computer program "RealTimer" (Open Science, Russia).

The *elevated plus maze* was performed to evaluate the level of anxiety in animals. The test was conducted in the cross-shaped maze, consisted of 2 closed (29 cm high walls) and 2 open (0.5 cm high sides) arms, and placed 100 cm above the floor. The size of arms was $52 \times 14 \text{ cm}$, with a central platform ($14 \times 14 \text{ cm}$) into the

arms crossing. The tested animal was placed into the central platform with its head toward the open arm. Behavioral parameters were registered during 5 min [48].

The *open-field test* was conducted to assess the level of anxiety, motor, and exploratory activity in animals. The test was conducted on the square arena (field size: $45 \text{ cm} \times 45 \text{ cm} \times 40 \text{ cm}$), the bottom of which was divided into 9 square sectors ($15 \text{ cm} \times 15 \text{ cm}$). The arena was illuminated with direct bright white light during the test and cleaned after the each animal. The rat was placed in the center of the arena; its behavior was registered for 5 min [43].

The *object recognition test* was performed to assess short-term object memory. A square arena ($45 \text{ cm} \times 45 \text{ cm} \times 40 \text{ cm}$) was used for the test. The behavioral test consisted of the three succeeding stages. On the first day of testing, the rat was placed in an empty arena for 5 min for habituation (this stage coincided with the open-field test). On the second day, the training and the testing stages were conducted. At the first stage (training), two objects of the same shape and size were placed in the arena. The rat was placed in the center of the arena, and the total exploration time (s) of objects was registered during 5 min. The second stage (test) was performed 2 h later. One of the objects was replaced by the new of different shape and size. The rat was placed in the center of the arena, and the total exploration time (s) of the familiar and new objects was registered. Animals that explored objects for less than 10 s during the training or the testing stages were excluded. The discrimination index for each animal was calculated according to the formula: $(T_n - T_o) / (T_n + T_o) \times 100$, where T_n is the total exploration time of the new object in the test stage, T_o is the total exploration time of the familiar object in the test stage [49, 50].

The *social interaction test* was performed to evaluate social activity. The test was conducted under the red light in the home cage of the testing rat, in which a juvenile rat of the same sex was placed for 10 min. The total time of social contacts (s) between the experimental rat and a juvenile rat was registered. Rats that exhibited aggressive behavior toward juvenile rats were excluded from the testing [40].

The *sucrose preference test* was conducted to assess the level of anhedonia in rats. Rats were placed in individual cages and presented with two bottles, one with pure water and the other with 1% sucrose solution for 1 day under free choice conditions. The bottles

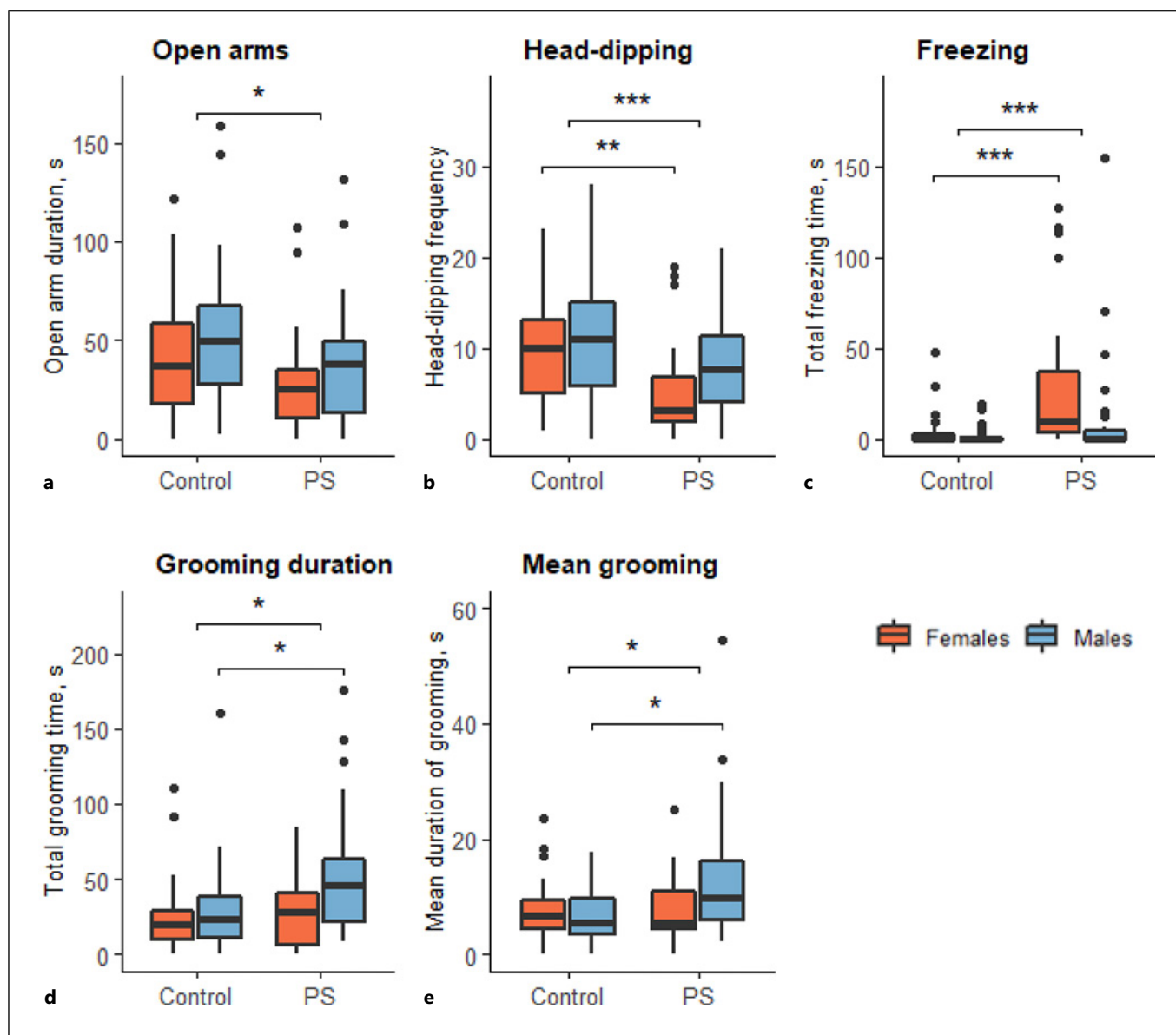


Fig. 2. a–e Elevated plus maze results. PS, offspring with prenatal stress; Control, control offspring; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

were swapped every 4 h to eliminate place preference. After 24 h, the volume of obtained fluid from both bottles was measured and the sucrose preference index was calculated according to the following formula: $V_s/(V_s + V_w) \times 100\%$, where V_w is the volume of drunk pure water, versus is the volume of obtained 1% sucrose solution [40].

The latent inhibition test was performed in an enclosed arena under dim light. The floor of the arena represented a grid with possibility of passage of current. At the first stage, the animals from each experimental group were divided into two subgroups: pre-exposure and non-pre-exposure. The animals were divided into subgroups independently from each other in terms of the total freezing time in the open-field test. Each animal from the pre-exposure subgroup was placed in the arena, and then a series of 40 sound signals (5 kHz) with a duration of 10 s and an interval of

50 s between the signals was applied. Animals from the non-pre-exposure subgroup were placed in the arena and kept there for 40 min without any sound exposure. The second stage was performed after 24 h. The animal was placed on the arena, and a sound signal (5 kHz, 10 s) was given, followed immediately by an electric current across the floor grid (10 mA, 1.5 s). This series of exposures was repeated again after 5 min. The third stage was performed 48 h after the second stage. The animal was placed in the arena, and after 30 s a sound signal (5 kHz) was given for 5 min. The total freezing time (s) of the animal was registered during the signal. The test results were analyzed to determine the differences in the total freezing time between pre-exposure and non-exposure subgroups in each experimental group. In healthy animals, prior exposure to a neutral stimulus (sound) should suppress its association with

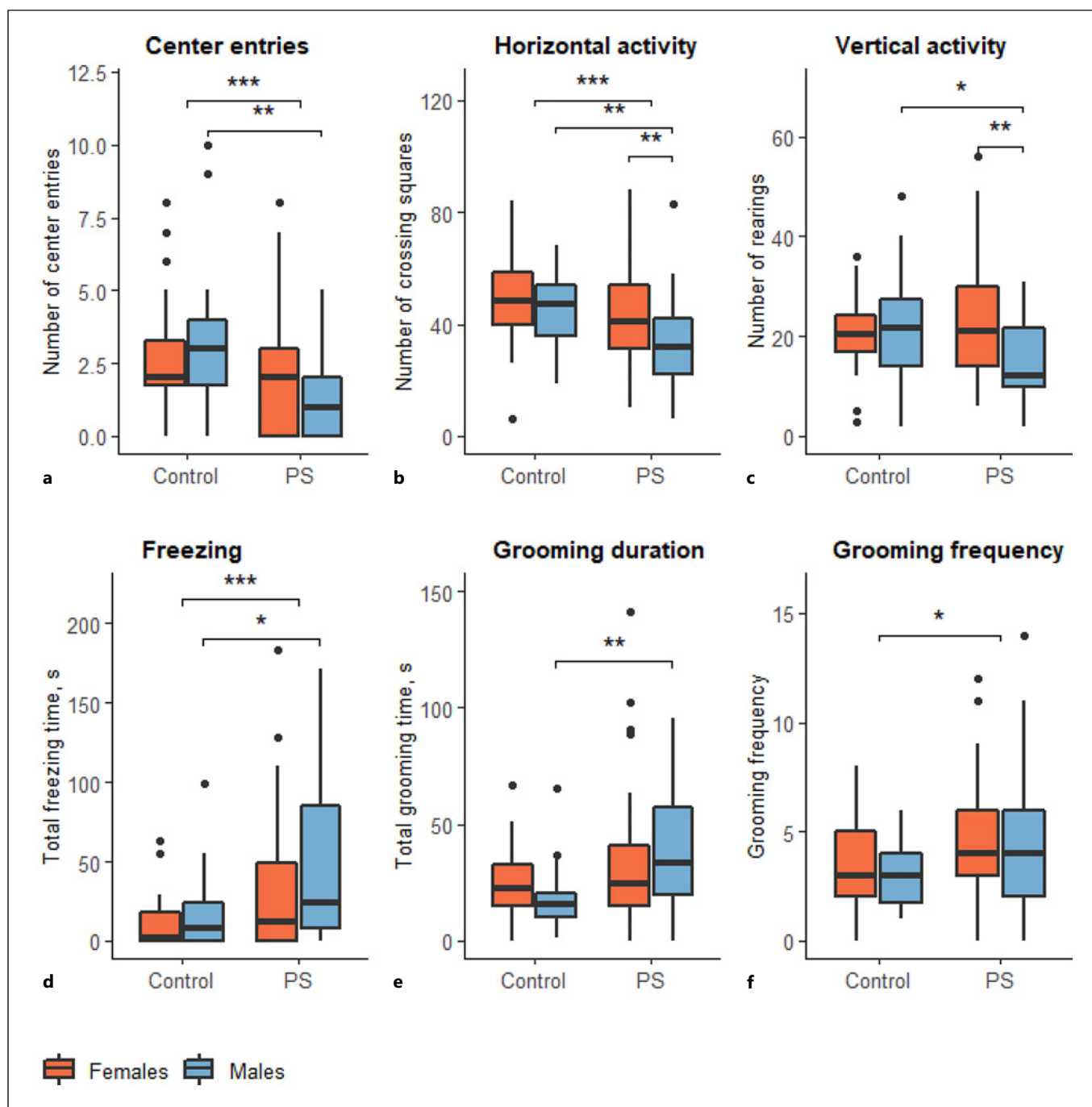


Fig. 3. a–f Results of the open-field test; PS, offspring with prenatal stress; Control, control offspring; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

exposure to electric current in the second stage. In the third stage, healthy animals should have a significant difference in the total freezing time between pre-exposure and non-exposure subgroups (pre-exposure animals demonstrate less freezing). Animals with impaired latent inhibition do not differ between subgroups [51, 52].

The *Morris water maze* was performed to evaluate long-term spatial memory. The maze represented a pool with gray walls

(diameter 150 cm), filled with water for 40 cm (temperature 23°C) and located in a room with a large number of spatial landmarks. A round platform 8 cm in diameter made of transparent plastic was placed in the center of one of the quadrants of the pool 2 cm below the water level. The training session consisted of 8 trials, during which the rat was placed into the different sectors of the pool and learned to find the platform. Platform search time (s) was

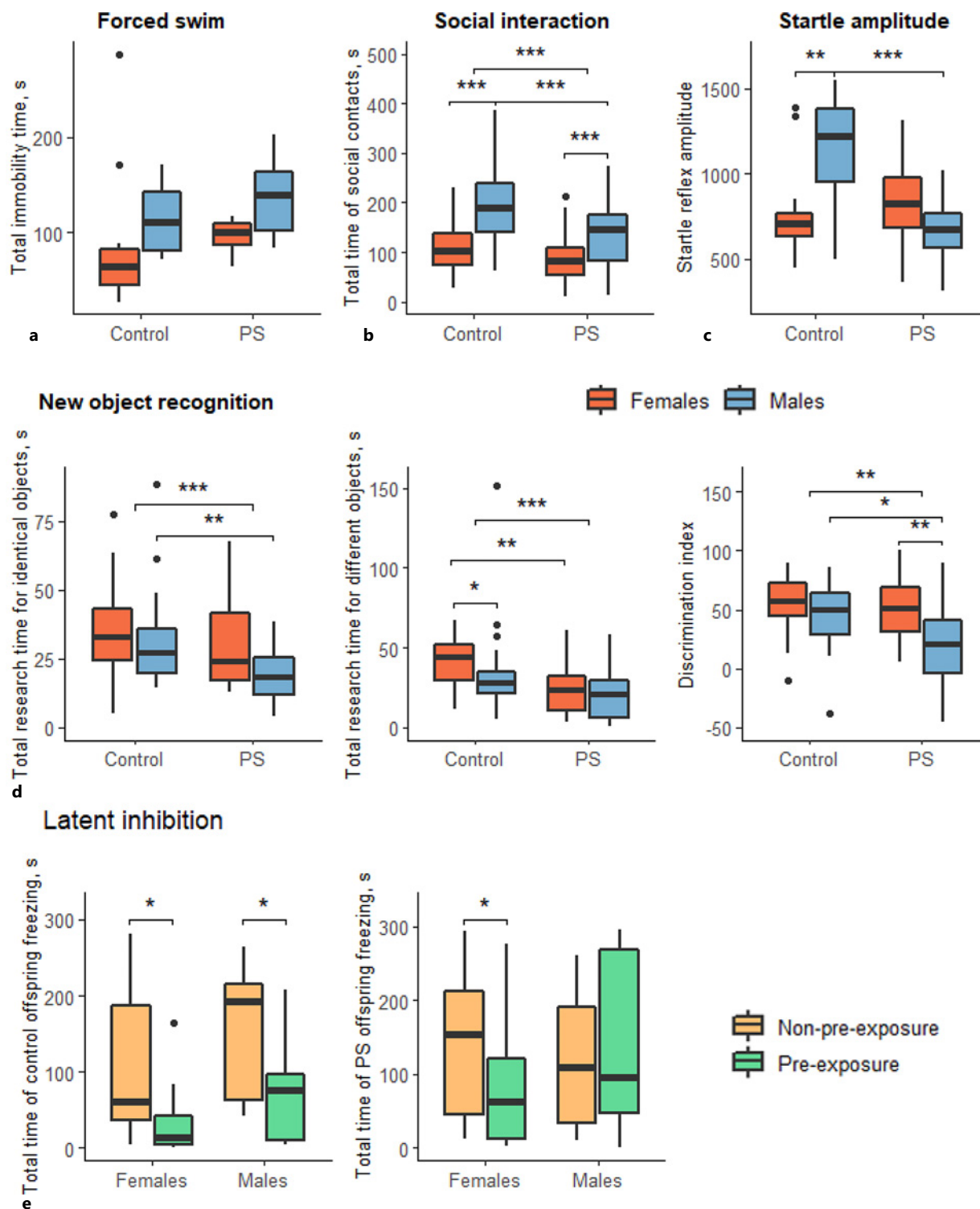


Fig. 4. Results of forced swimming test (a), social interaction test (b), start reflex test (c), object recognition test (d), latent inhibition test (e); PS, offspring with prenatal stress; Control, control offspring; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Table 1. Results of correlation analysis of behavioral test scores

Behavioral indicator 1	Behavioral indicator 2	R	p value
Control males			
Head-dipping frequency (EPM)	Total time of social contacts (SI)	−0.60	0.003
Total grooming time (OF)	Total time of social contacts (SI)	−0.64	0.001
Frequency of grooming (OF)	Total time of social contacts (SI)	−0.67	<0.001
Mean duration of grooming (OF)	Total research time for different objects (OR)	−0.49	0.006
Total grooming time (OF)	Sucrose preference index (A)	−0.46	0.008
PS males			
Frequency of crossing the center platform (EPM)	Total research time for identical objects (OR)	0.54	0.003
Frequency of looking out of closed arms (EPM)	Total research time for different objects (OR)	0.66	<0.001
Total grooming time (EPM)	Total research time for identical objects (OR)	−0.56	0.002
Total freezing time (OF)	Total research time for different objects (OR)	−0.62	0.008
Mean duration of grooming (EPM)	Total research time for different objects (OR)	−0.62	<0.001
Total grooming time (OF)	Sucrose preference index (A)	0.63	0.007
Mean duration of grooming (OF)	Sucrose preference index (A)	0.69	0.002
Control females			
Head-dipping frequency (EPM)	Total time of social contacts (SI)	−0.54	0.009
Total time spent in open arms (EPM)	Total research time for different objects (OR)	0.51	0.004
Head-dipping frequency (EPM)	Total research time for different objects (OR)	0.50	0.004
PS females			
Head-dipping frequency (EPM)	Total time of social contacts (SI)	0.62	0.006
Frequency of grooming (EPM)	Total time of social contacts (SI)	0.71	<0.001
Frequency of rearing (EPM)	Discrimination index (OR)	0.59	0.007
Head-dipping frequency (EPM)	Sucrose preference index (A)	0.78	<0.001
Frequency of rearing (EPM)	Sucrose preference index (A)	0.66	<0.001
Total time of social contacts (SI)	Sucrose preference index (A)	0.81	<0.001

A, sucrose preference test; OF, open-field test; EPM, elevated plus maze; PS, prenatal stress; OR, object recognition test; SI, social interaction test.

registered. Animals that did not find the platform within 60 s were gently guided to it. The animal was left on the platform for 15 s, and then placed in an individual cage for 60 s. The test was conducted after 48 h. The animal was placed in the pool, and platform search time (s) was registered [43, 53].

The forced swimming test was performed to assess despair behavior. The test was performed in a transparent cylinder (diameter 31 cm, height 40 cm) filled with water (temperature 23°C) to a height of 30 cm from the bottom of the cylinder. The animal was placed in a cylinder for 8 min. The number of diving, active swimming time (s), passive swimming time (s), and immobility time (s) were registered during the last 6 min of the test [54, 55].

Evaluation of acoustic startle reflex (ASR) and prepulse inhibition: the prepulse inhibition test was performed using the TSE-systems' experimental startle reaction unit (TSE, Bad Homburg, Germany). The test was performed according to the previously given protocol [56], but with slight modifications. The rats were tested in soundproof boxes inside confining metal cages, which were installed on a platform (27 × 9 × 19 cm) equipped with a pressure sensor. The acoustic stimulus was presented through two loudspeakers located along the walls of metal cages. All animals were presented with 18 sound stimuli after a 5-min adaptation. The first four stimuli were presented without a prepulse. The following 10 sound signals were presented in a pseudorandom order, including single-pulse

presentations and five presentations with a prepulse 100 ms before the main sound signal. The remaining 4 sound signals were presented without a prepulse. The intervals between the sound signals were 15 s. The main stimulus used was broadband noise with a duration of 20 ms and a volume of 105 dB. The background masking sound signal had a broadband noise of 72 dB, and the prepulse had a signal duration of 20 ms and a volume of 80 dB. The magnitude of the prepulse inhibition was calculated according the formula: $Ap/Am \times 100\%$, where Am is the mean amplitude of the startle response without a prepulse, excluding the first and last four, and Ap is the mean amplitude of the startle response combined with the prepulse.

Collection and Storage of Biomaterials

Rat brain samples for biochemical analysis were collected after brief anesthesia followed by rapid decapitation. The brain sections (frontal cortex and hippocampus) were isolated after the brain extraction. Brain samples immediately after the isolation were placed into cryoprobes and frozen in liquid nitrogen, where they stored there until the analysis.

Determination of Neurotransmitter Concentrations in the Brain

The methods most used to measure neurotransmitters include electrochemical methods using micro- and nanoelectrodes, biosensors, liquid chromatography, and mass spectrometry [57–59].

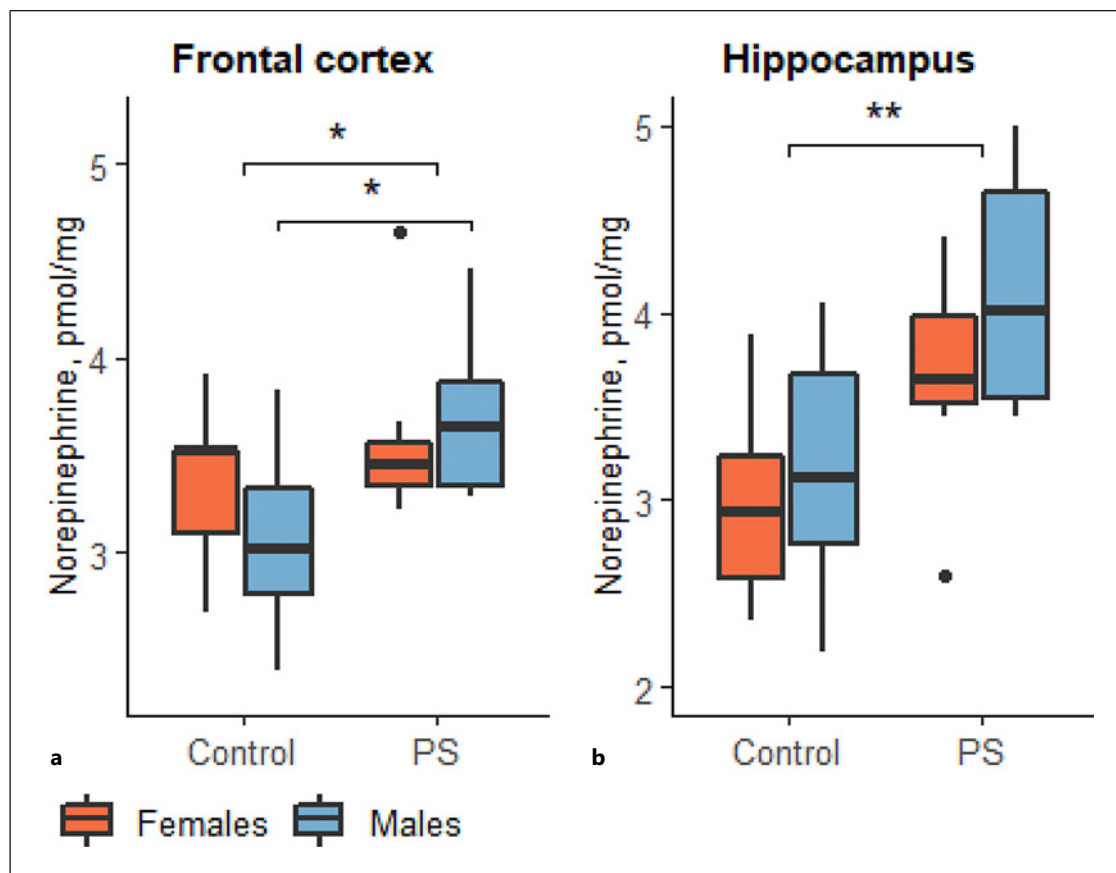


Fig. 5. Norepinephrine concentration in the frontal cortex (a) and hippocampus (b) of rats. PS, offspring with prenatal stress; Control, control offspring; * $p < 0.05$; ** $p < 0.01$.

In this work, we used the HPLC-ED method as one of the standard methods for detection of neurotransmitters. In the brain tissue samples (frontal cortex and hippocampus), the concentration of norepinephrine, dopamine, 3,4-dihydroxyphenylacetic acid (DOPAC), 5-hydroxyindoleacetic acid (HIAA) and serotonin (below analytes) were determined. The samples were homogenized in 0.1 n HClO₄ (Sigma Aldrich, USA) with 50 pmol/mL internal standard 3,4-dihydroxybenzylamine hydrobromide (Sigma Aldrich, USA) via the ultrasonic homogenizer (Sartorius, France). After that, the solution was centrifuged for 15 min at 15,000 rpm. HPLC separation was carried out on the reversed-phase column ReproSil-Pur, ODS-3, 4 × 100 mm with a pore diameter of 3 μm (Dr. Majsch, Germany) at +28°C and a mobile phase speed of 1 mL/min supported by the liquid chromatograph LC-20ADsp (Shimadzu, Japan). The mobile phase consisted of 0.1 M citrate-phosphate buffer, 0.3 mM sodium octanesulfonate, 0.1 mM EDTA, and 8% acetonitrile (all reagents from Sigma Aldrich, USA), pH 2.58. The electrochemical detector DECADE II (Antec Leyden, The Netherlands) was equipped with a working glassy carbon electrode (+0.80 V) and an Ag/AgCl reference electrode. The peaks of interest and internal standard were identified by their release time in the standard solution. The analytes concentration was calculated by

the internal standard method using a calibration curve with LabSolutions software (Shimadzu, Japan). Samples were normalized by tissue weight [60]. Based on the data obtained, the DOPAC/dopamine and HIAA/serotonin ratios were calculated as indicators of dopamine and serotonin metabolic turnover, respectively [61].

Statistics

Statistical analysis of the obtained data and plotting were performed using RStudio (Version 1.4, RStudio PBC, USA), jamovi (Version 1.6, the jamovi project, Australia), and STATISTICA (Version 12, StatSoft, USA) software. The normality of the data distribution was estimated using the Shapiro-Wilk criterion. In the case of a normal distribution, p values were calculated using ANOVA followed by Tukey test for multiple comparisons; the results were presented as mean ± SEM. In the case of nonnormal distribution, nonparametric Kruskal-Wallis criterion, and Mann-Whitney U-criterion with Dwass-Steel-Critchlow-Fligner for multiple comparisons were used; results were presented as Med (Q1; Q3). p value <0.05 was considered as statistically significant.

ANOVA with repeated measures followed by Tukey test for multiple comparisons was used to assess repeated

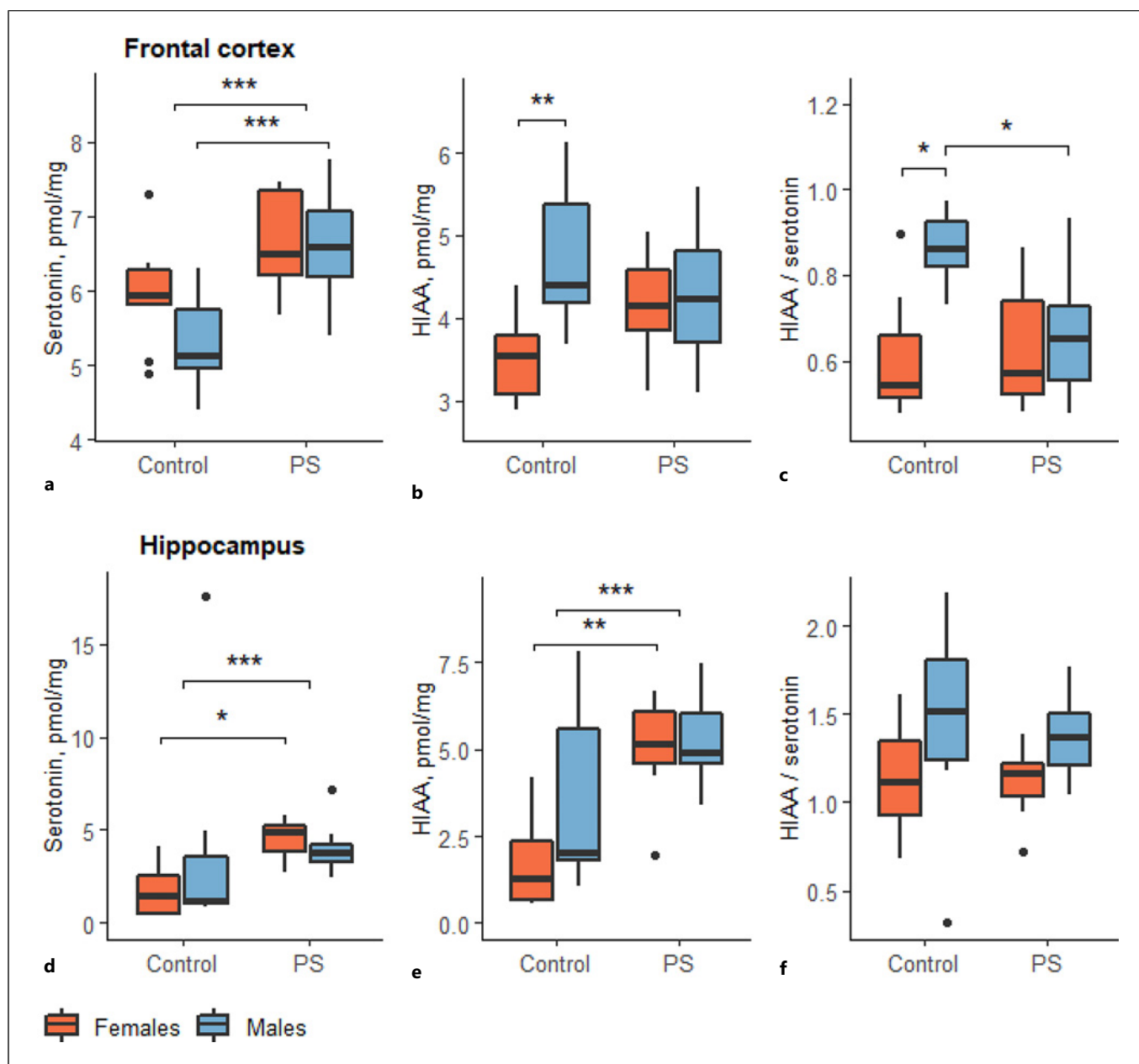


Fig. 6. Results of determination of indicators of serotonergic metabolism in the frontal cortex (a–c) and hippocampus (d–f) of rats. PS, offspring with prenatal stress; Control, control offspring; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

measures (Morris water maze training). p value < 0.05 was considered as statistically significant.

Analysis of covariance (ANCOVA), followed by Tukey test for multiple comparisons, was used to determine the results of the latent inhibition test. The total freezing time in the open-field test was taken as the covariate in this case. p value < 0.05 was considered statistically significant.

Spearman correlation was performed for the correlation analysis. $p < 0.01$ was considered statistically significant for the correlation analysis to account for multiple comparisons.

Results

In our research, we studied the effect of US PS on the behavioral phenotype of rats (anxiety and exploratory behavior, social behavior, cognitive functions, depressive-like behavior, startle reflex, and latent inhibition) and also evaluated the correlation between the behavioral parameters and neurotransmitter systems of the brain.

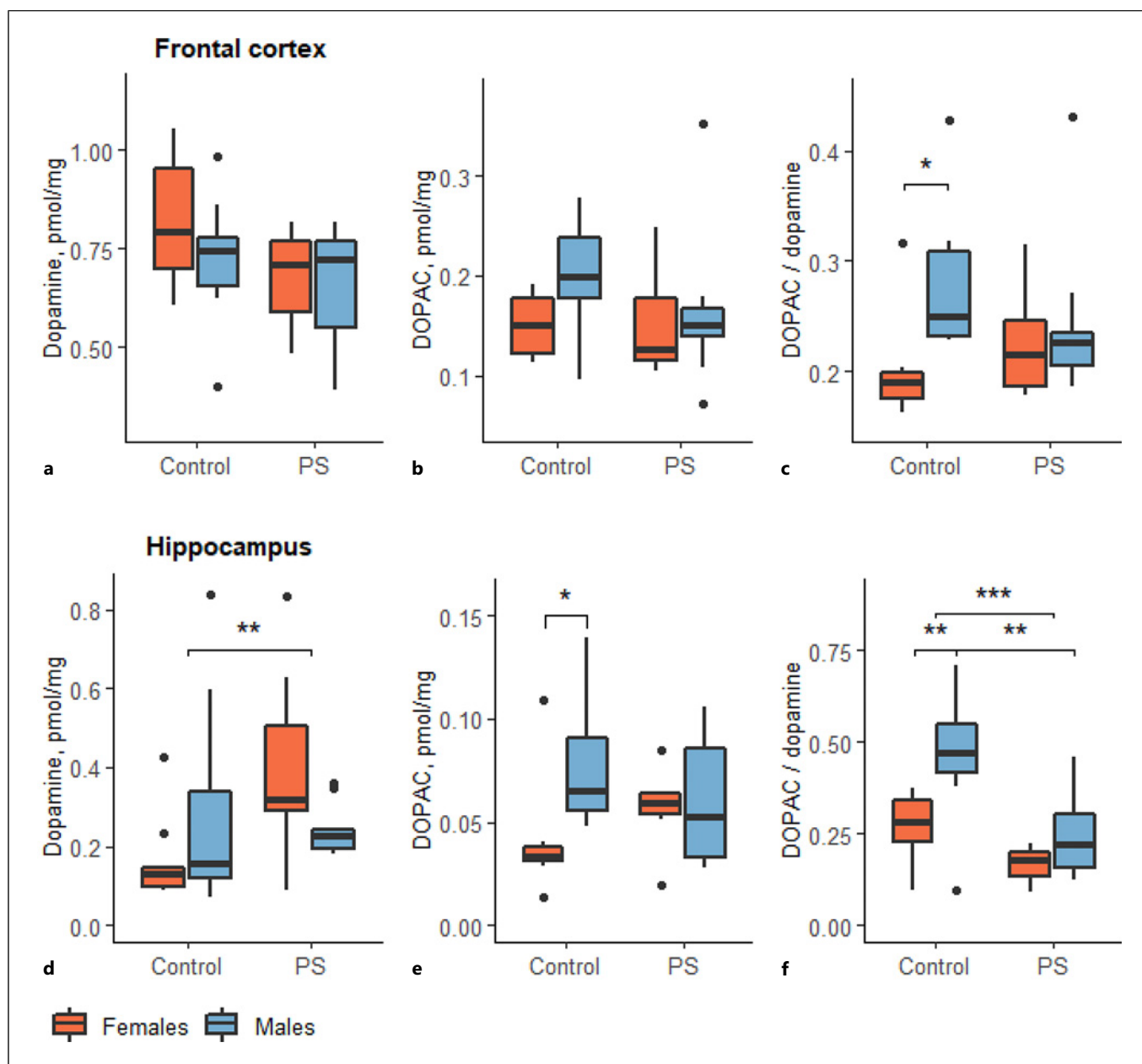


Fig. 7. Results of determination of parameters of dopaminergic metabolism in the frontal cortex (a–c) and hippocampus (d–f) of rats. PS, offspring with prenatal stress; Control, control offspring; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Evaluation of Anxiety and Exploratory Behavior Elevated plus Maze

PS significantly influenced the rat's behavior in this test. Statistically significant differences were demonstrated for the total time spent in open arms ($\chi^2 = 10.28$; $p = 0.02$). PS offspring spent less time in open arms compared to controls ($p = 0.01$) (shown in Fig. 2a). However, a posteriori analysis revealed no significant differences.

Kruskal-Wallis test demonstrated significant differences between the groups by head-dipping frequency ($\chi^2 = 15.59$; $p = 0.001$) and total freezing time ($\chi^2 = 29.70$; $p < 0.001$). Statistically significant differences were revealed between the controls and PS females (shown in Fig. 2b, c). PS females demonstrated decreased head-dipping frequency ($p = 0.007$) and increased freezing ($p < 0.001$) compared to the control females. In addition, PS

Table 2. Correlation analysis of the relationship between behavior and concentrations of biochemical indicators of brain tissue

Group	Biochemical indicator	Behavioral indicator	<i>R</i>	<i>p</i> value
Serotonergic system				
PS offspring	HIAA (FC)	Frequency of diving (FS)	−0.63	0.007
	HIAA/serotonin ratio (FC)	Frequency of diving (FS)	−0.61	0.0097
Control males	HIAA (FC)	Horizontal activity (OF)	−0.87	0.004
	HIAA (FC)	Number of center entries (OF)	−0.82	0.007
	HIAA (H)	Total grooming time (EPM)	0.85	0.006
PS males	HIAA (FC)	Total grooming time (OF)	−0.78	0.008
	HIAA (FC)	Frequency of diving (FS)	−0.79	0.006
PS females	Serotonin (FC)	Total active swimming time (FS)	−0.96	0.003
Dopaminergic system				
Control offspring	DOPAC/dopamine ratio (H)	Frequency of crossing the center platform (EPM)	0.59	0.008
	DOPAC/dopamine ratio (H)	Total freezing time (EPM)	−0.60	0.006
PS offspring	DOPAC/dopamine ratio (FC)	Head-dipping frequency (EPM)	−0.65	0.004
Control males	DOPAC (FC)	Frequency of rearing (OF)	−0.89	0.001
PS males	Dopamine (FC)	Discrimination index (OR)	−0.79	0.006
PS females	DOPAC (FC)	Mean duration of grooming (EPM)	−0.88	0.008
Norepinephrine				
Control offspring	Norepinephrine (FC)	Total freezing time (OF)	0.590	0.0099
PS offspring	Norepinephrine (FC)	Discrimination index (OR)	−0.77	0.003
	Norepinephrine (H)	Total passive swimming time (FS)	−0.67	0.004
PS females	Norepinephrine (FC)	Frequency of rearing (OF)	−0.92	0.003

H, hippocampus; OF, open-field test; EPM, elevated plus maze; FS, forced swimming test; PS, prenatal stress; OR, object recognition test; FC, frontal cortex.

females showed more prolonged freezing than PS males ($p = 0.005$); the control group have not demonstrated this difference.

Kruskal-Wallis test also revealed significant differences in total grooming time ($\chi^2 = 12.98$; $p = 0.005$) and mean duration of grooming ($\chi^2 = 9.41$; $p = 0.02$). However, significant differences were observed only in males (shown in Fig. 2d, e). PS males showed increased grooming time ($p = 0.03$) and increased mean duration of grooming ($p = 0.02$) compared to the control males. In addition, the effect of the sex factor was also demonstrated for the total grooming time: males groomed longer than females ($p = 0.02$). Experimental groups did not differ significantly by the frequency of grooming ($\chi^2 = 4.31$; $p = 0.23$), number of peeking ($\chi^2 = 7.04$; $p = 0.07$), frequency of the center crossing ($\chi^2 = 2.45$; $p = 0.48$), and number of rearing ($\chi^2 = 7.37$; $p = 0.06$).

Open-Field Test

Overall statistically significant differences were found for the following parameters: number of center entries ($\chi^2 = 16.91$; $p < 0.001$), horizontal activity ($F = 8.81$; $p < 0.001$), vertical activity ($\chi^2 = 16.27$; $p = 0.001$), total

freezing time ($\chi^2 = 17.90$; $p < 0.001$), total grooming time ($\chi^2 = 15.46$; $p = 0.001$), and grooming frequency ($\chi^2 = 7.86$; $p = 0.049$). At the same time, analysis of posterior comparisons demonstrated statistically significant differences only between the control and PS males. PS males demonstrated decreased number of the center entries ($p = 0.001$) (shown in Fig. 3a), decreased horizontal activity ($p = 0.004$) (shown in Fig. 3b), reduced vertical activity ($p = 0.02$) (shown in Fig. 3c), and increased freezing ($p = 0.02$) (shown in Fig. 3d) and grooming time ($p = 0.001$) (shown in Fig. 3e) compared to the control males. In addition, PS offspring demonstrated increased number of grooming compared to the controls ($p = 0.01$) (shown in Fig. 3f), but no differences were shown separately for males and females. Control and PS females did not differ significantly from each other. The sex factor influenced the behavior of the PS offspring – PS males showed less horizontal activity compared to PS females ($p = 0.008$), while no significant differences by the horizontal activity parameter were revealed between the control females and males ($p = 0.56$) (shown in Fig. 3b). The similar pattern was observed for vertical activity – PS males demonstrated decreased number of rearings compared to the PS

females ($p = 0.002$), while the control males and females did not differ significantly by the number of rearings ($p = 1.00$) (shown in Fig. 3c).

Evaluation of Depressive-Like Behavior

Sucrose Preference Test

PS produced no significant effect on the level of anhedonia in rats since the sucrose preference index did not differ significantly between the groups ($F = 0.41$; $p = 0.75$).

Forced Swimming Test

Kruskal-Wallis test demonstrated statistically significant differences only for the total immobility time ($\chi^2 = 10.00$; $p = 0.02$) (shown in Fig. 4a). Nevertheless, the posterior comparisons analysis revealed only significant overall differences between the sexes for this parameter: males freeze longer than females ($p = 0.005$). However, males and females did not differ separately in the control and PS groups. Immobility time did not differ significantly in the controls and PS offspring ($p = 0.09$). No significant overall differences were demonstrated for the active swimming time ($F = 2.69$; $p = 0.062$), passive swimming time ($F = 0.66$; $p = 0.58$), and number of diving ($\chi^2 = 5.46$; $p = 0.14$).

Evaluation of Social Behavior

PS altered social behavior in adult rat offspring in the social interaction test. Overall statistically significant differences were found for the total time of social contacts ($F = 28.79$; $p < 0.001$). PS offspring demonstrated decreased social interaction time with juvenile animals compared to the control offspring ($p < 0.001$). Nevertheless, the differences was observed only in males ($p < 0.001$), control, and PS females did not differ by the level of social activity ($p = 0.34$) (shown in Fig. 4b). The test results also revealed statistical differences between the sexes by the social interaction time: males contacted juvenile animals longer than females ($p < 0.001$); it was observed in both control ($p < 0.001$) and PS ($p < 0.001$) offspring.

Evaluation of Cognitive Functions

Object Recognition Test

Kruskal-Wallis test revealed the significant overall differences in the total exploration time of the identical objects ($\chi^2 = 19.6$; $p < 0.001$) and in the total exploration time of the different objects (test phase) ($\chi^2 = 22.8$; $p < 0.001$) (shown in Fig. 4d). The total exploration time of the identical objects was reduced in PS males compared to the control males ($p = 0.006$), but no significant differences were observed in the control and PS females ($p = 0.66$). At the same time, the total exploration time of

different objects did not differ significantly in the control and PS males ($p = 0.10$), but it was decreased in PS females compared to the control females ($p = 0.002$). The test also revealed the differences between the sexes in the control group: males demonstrated decreased exploration time of different objects compared to the females ($p = 0.03$), which was not observed in the PS groups ($p = 0.77$).

ANOVA revealed statistically significant differences of the discrimination index ($F = 6.84$; $p < 0.001$). Discrimination index was reduced in PS offspring compared to the control offspring ($p = 0.009$). A posteriori comparisons analysis showed that PS decreased the discrimination index in males ($p = 0.02$), but not in females ($p = 0.88$). At the same time, control males and females did not differ from each other by the discrimination index ($p = 0.38$), and the male discrimination index was significantly decreased compared to the females of the PS group ($p = 0.005$) (shown in Fig. 4d).

Morris Water Maze

The training stage was evaluated using ANOVA with repeated measures, which allows to estimate the overall effect of the factor of repeated measures (number of trials) ($F = 57.34$; $p < 0.001$). Platform search time decreased from the first to the fourth trials (for trials 1 and 2 $p < 0.001$; for trials 2 and 3 $p = 0.005$; for trials 3 and 4 $p = 0.03$); no significant differences by the platform search time were observed from the fifth to the eighth trials. The analysis revealed no interaction of the repetitive change factor with the experimental group factor ($F = 0.74$; $p = 0.64$) and with gender factor ($F = 0.38$; $p = 0.91$). The test also demonstrated no overall differences in learning between the PS and control groups ($F = 2.13$; $p = 0.15$), or between males and females ($F = 2.88$; $p = 0.09$). Test results 48 h after the training also showed no significant differences between the groups ($\chi^2 = 0.19$; $p = 0.98$).

ASR and Prepulse Inhibition

Evaluation of ASR in adult offspring showed significant overall statistical differences in startle reflex amplitude ($F = 6.48$; $p < 0.001$). Amplitude was significantly lower in PS males compared to the control males ($p < 0.001$) (shown in Fig. 4c). Control and PS females did not differ statistically ($p = 0.84$). In addition, differences were detected between the sexes in the control group: the amplitude in the control males was higher than that in the control females ($p = 0.007$). The sex factor produced no effect on the amplitude in PS offspring ($p = 0.35$). ANOVA also revealed no significant differences in the level of prepulse inhibition among the experimental groups ($F = 1.25$; $p = 0.30$).

Latent Inhibition

ANCOVA revealed overall significant differences between the groups ($F = 3.35$; $p = 0.002$). The influence of the pre-exposure factor was also detected ($F = 11.07$; $p = 0.001$). The Mann-Whitney test was used to evaluate the effect of the pre-exposure stimulus because of the non-normal data distribution. Analysis was performed separately for each experimental group since baseline anxiety (total freezing time in the open-field test) differed significantly between the groups. Control males after the pre-exposure stimulus demonstrated decreased freezing time in the third stage of the test compared to the control males without the pre-exposure stimulus ($U = 29$; $p = 0.04$), indicating that the control males express a normal latent inhibition (shown in Fig. 4e). Control females showed the similar results: control females froze significantly less after the pre-exposure stimulus compared to the control females without the pre-exposure stimulus ($U = 27$; $p = 0.03$). PS offspring demonstrated other results (shown in Fig. 4e). The pre-exposure stimulus factor produced no effect on PS males ($U = 61$; $p = 0.78$), indicating impaired latent inhibition. However, the PS females demonstrated decreased freezing time after the pre-exposure stimulus compared to the PS females without the pre-exposure stimulus ($U = 27$; $p = 0.02$), indicating normal latent inhibition.

Correlation Analysis of Behavioral Test Scores

A correlation analysis was performed to evaluate the association of behavioral phenotype traits that demonstrated the multiple correlations of anxiety indicators with other behavioral parameters. The significant results of the analysis ($p < 0.01$) are presented in Table 1.

Control males demonstrated the correlations between grooming indicators in the open-field test, traits of social behavior; anhedonia level, and duration of object exploring in the object recognition test. PS males showed multiple correlations between the open field and EPM test parameters and the exploration time in the object recognition test. In addition, the certain correlation between grooming and anhedonia level was also indicated in PS males. Control females demonstrated the correlations of the EPM test parameters with the exploration time in the object recognition test and with the social interaction time. At the same time, PS females showed the correlations of EPM and open-field parameters scores with social behavior traits and anhedonia level.

All experimental groups of animals, except PS male group, demonstrated multiple correlations between the indicators of social behavior and anxiety. In particular, the inverse correlation was found between social and

anxious behavioral traits in the control males; animals with increased anxiety were characterized by a reduced level of social contacts. Social behavior was also negatively correlated with intensity of grooming behavior, which can possibly indicate socially significant replacement behavior. At the same time, social behavior was positively correlated with dangling in EPM, the indicator of risk-assessment behavior. In contrast to the control males, PS males demonstrated no correlation of anxious behavior with social activity. Nevertheless, this group was characterized by multiple correlations of anxiety behavior and object exploration activity in the object recognition test. Based on this, we hypothesize that impaired social behavior is not strongly related to the level of anxiety in PS males, oppositely to the controls. However, the anxious behavior of PS males is associated with the exploration of nonliving objects. In PS females, the correlation of social activity and anxious behavior was detected in the same way as in the control groups, but it did not represent a unidirectional form.

Determination of Norepinephrine Concentration in the Rat Brain Structures

Kruskal-Wallis test revealed significant overall differences between the groups by the noradrenaline concentration in the frontal cortex ($\chi^2 = 8.65$; $p = 0.03$). PS increased norepinephrine levels in the offspring ($p = 0.03$). However, analysis of pairwise comparisons showed the increase of norepinephrine concentration only in PS males compared to the control males ($p = 0.04$) (shown in Fig. 5a). Control and PS females did not differ significantly by the noradrenaline concentration in the frontal cortex ($p = 0.99$).

ANOVA showed significant differences between the groups in hippocampus norepinephrine concentration ($F = 5.53$; $p = 0.004$). Norepinephrine concentration was higher in the hippocampus of the PS offspring compared to the control offspring ($p = 0.003$) (shown in Fig. 5b). However, posterior analysis demonstrated no significant differences between PS and control offspring separately for males ($p = 0.06$) and females ($p = 0.21$). Additionally, the statistical differences were detected between the sexes at the level of significance ($p = 0.05$): males demonstrated the higher concentrations of noradrenaline compared to the females.

Determination of Serotonergic Metabolism in the Rat Brain Structures

ANOVA revealed significant differences between the groups by the serotonin concentration in the frontal cortex ($F = 8.38$; $p < 0.001$) (shown in Fig. 6a). PS

offspring were characterized by elevated serotonin levels compared to the control offspring ($p < 0.001$). Nevertheless, analysis of posterior comparisons showed the presence of differences only in males ($p < 0.001$), but not in females ($p = 0.19$).

ANOVA demonstrated significant differences between the groups by HIAA concentration in the frontal cortex ($F = 3.94$; $p = 0.02$) (shown in Fig. 6b). The control and PS offspring did not differ significantly between each other ($p = 0.72$). However, statistical differences were indicated between the sexes ($p = 0.02$). A posteriori comparisons analysis revealed that control males demonstrated significantly higher HIAA concentrations compared to the control females ($p = 0.01$). PS effect eliminated these differences ($p = 0.99$). In addition, the interaction of group factor and gender factor has been detected ($F = 4.57$; $p = 0.04$).

Kruskal-Wallis test revealed significant differences between the groups by HIAA/serotonin ratio in the frontal cortex ($\chi^2 = 12.6$; $p = 0.006$) (shown in Fig. 6c). The ratio was statistically higher in the control males compared to the control females ($p = 0.02$), which was not observed in PS offspring ($p = 1.00$). In addition, the HIAA/serotonin ratio was increased generally in males compared to the females ($p = 0.02$). The HIAA/serotonin ratio was lower in PS males compared to the control males ($p = 0.03$), but the females did not differ by this parameter ($p = 0.96$).

Kruskal-Wallis test revealed significant differences between the groups by the hippocampal serotonin concentration ($\chi^2 = 13.35$; $p = 0.004$) (shown in Fig. 6d). PS offspring were characterized by the increased concentration of serotonin compared to the control offspring ($p < 0.001$). Analysis of pairwise comparisons showed that only PS females demonstrated elevated serotonin concentrations compared to the control females ($p = 0.02$); the males did not differ by that parameter ($p = 0.24$).

Kruskal-Wallis test also detected statistical differences between the groups by the hippocampal HIAA concentration ($\chi^2 = 15.89$; $p = 0.001$) (shown in Fig. 6e). PS offspring demonstrated the increased concentration of HIAA compared to the controls ($p < 0.001$). However, analysis of pairwise comparisons showed that only PS females had increased serotonin concentrations compared to the control females ($p = 0.01$), males did not differ significantly by this parameter ($p = 0.36$). ANOVA revealed no significant overall differences between the groups by the HIAA/serotonin hippocampal ratio ($F = 2.18$; $p = 0.11$) (shown in Fig. 6f).

Determination of Dopaminergic Metabolism in the Rat Brain Structures

PS did not alter dopamine ($\chi^2 = 4.34$; $p = 0.23$) or DOPAC ($\chi^2 = 5.23$; $p = 0.16$) concentration in the frontal cortex. However, Kruskal-Wallis test revealed significant differences in the DOPAC/dopamine ratio ($\chi^2 = 12.2$; $p = 0.007$) (shown in Fig. 7a–c). Analysis of pairwise comparisons showed statistical differences between the sexes in the control group; the DOPAC/dopamine ratio was higher in control males compared to the control females ($p = 0.02$). No significant differences between the sexes were observed in the PS offspring.

Kruskal-Wallis test revealed statistical differences between the groups by the hippocampal dopamine concentration ($\chi^2 = 8.75$; $p = 0.03$) (shown in Fig. 7d). PS offspring demonstrated the increased dopamine concentration compared to the control offspring ($p = 0.009$). However, analysis of pairwise comparisons showed that males ($p = 0.51$) and females ($p = 0.14$) did not differ separately.

Kruskal-Wallis test also detected significant differences between the groups by the hippocampal DOPAC concentration ($\chi^2 = 9.84$; $p = 0.02$) (shown in Fig. 7e). Analysis of pairwise comparisons revealed that control males demonstrated the increased concentration of DOPAC compared to the control females ($p = 0.01$). Nevertheless, PS offspring showed no statistical differences between the sexes ($p = 1.00$). PS also produced no effect on DOPAC concentration in the hippocampus.

ANOVA showed significant differences in the hippocampal DOPAC/dopamine ratio ($F = 9.75$; $p < 0.001$) (shown in Fig. 7f). PS reduced the DOPAC/dopamine ratio in the hippocampus ($p < 0.001$). A posteriori analysis revealed significant differences only between the control and PS males ($p = 0.002$), but not between the females ($p = 0.32$). In addition, significant differences were detected between the sexes. Females demonstrated the lower DOPAC/dopamine ratio compared to the males ($p = 0.001$). However, posterior analysis showed significant differences between the sexes only in the control offspring ($p = 0.005$), but not in the PS offspring ($p = 0.49$).

Correlation Analysis of Behavioral and Biochemical Parameters

Correlation analysis was performed to evaluate the relationship between the elements of behavioral phenotype and biochemical parameters of the brain. The analysis demonstrated the presence of multiple correlations between biochemical and behavioral parameters.

The significant results of the analysis ($p < 0.01$) are presented in Table 2.

Correlation analysis demonstrated that behavioral traits are related to the parameters of the serotonergic system, mainly in the frontal cortex. Elevated levels of HIAA metabolite and a high HIAA/serotonin ratio (i.e., high levels of serotonin metabolism) were associated with negative behavioral traits: decreased locomotor activity, increased anxiety, active grooming in the control group, and also with decreased activity in the forced swim test in the PS group. Elevated levels of HIAA in the frontal cortex were associated with less active grooming in PS males, whereas increased levels of HIAA in the hippocampus were associated with more active grooming in control males.

The character of correlations of the dopaminergic system and animals behavior differed in the hippocampus and frontal cortex. Elevated DOPAC/dopamine ratio (i.e., the index of high dopamine metabolism) in the hippocampus of control animals was associated with decreased anxiety. However, no associations were found in the PS offspring hippocampus. At the same time, the increased dopamine metabolism in the frontal cortex of control and PS groups was associated with pathological behavior: reduced exploration activity, and impaired cognitive function.

Increased frontal cortex noradrenaline concentration was associated with negative behavioral traits: impaired cognitive function, decreased exploration activity, and increased anxiety in the control and PS offspring. However, the high norepinephrine concentration in the PS offspring hippocampus was associated with decreased passive swimming time in the forced swimming test.

Therefore, we can suggest that US PS altered the behavioral phenotype of the offspring: US PS increased the level of anxious behavior, impaired exploratory behavior, and increased grooming activity. Social behavior was also altered: males demonstrated decreased social contacts, impaired latent inhibition, and decreased ASR levels. Their behavioral reactions shifted from social interaction to interaction with the nonliving objects. The second part of the experiment demonstrated the effect of US PS on neurotransmitter systems of the brain. Our study demonstrated significant results for the serotonergic, dopaminergic, and noradrenergic neurotransmitter systems in the frontal cortex and hippocampus. The US PS activated all of these systems, increasing the concentration of neurotransmitters in the brain. The presence of associations between biochemical and behavioral indicators in our study supports the suggestion of their interrelation.

Discussion

Our study revealed that US PS affected some behavioral traits and biochemical parameters of the adult rat brain. Below, we will discuss the observed alterations in more detailed way.

Changes of the Rat Behavioral Phenotype under the US PS Exposure

We observed impaired exploratory behavior, the presence of anxious traits and increased passive-defensive reactions of adult rats after the US PS in the open-field test and EPM. The results of our study correspond to the data of numerous research studies, devoted to the study of PS action on animal behavior. Different types of PS cause anxious behavior in animals of different ages [62–66]. In addition, we also previously showed similar results in the open-field test for the juvenile rat offspring after the exposure to US PS [46]. Another aspect of behavior that we studied in this experiment is grooming. US PS increased grooming activity in male rats. It has previously been shown that stress can enhance grooming in rodents [67–69]. Our study also indicated this effect for PS.

The important indicator of social activity is the degree of interest to unfamiliar individuals of one's species, i.e., the level of preference for social novelty [70]. This behavioral aspect was studied using the social interaction test, which showed a negative effect of US PS on the male's social contacts. Our results are supported by the earlier studies, which have demonstrated that different types of PS produce a negative effect on the animal social activity [64, 71–73].

The present data addressed to the effect of PS on latent inhibition are few and contradictory since different types of PS produce opposite effects on latent inhibition in animals. For example, the administration of dexamethasone to female rats during pregnancy does not impair latent inhibition in the offspring [74]. Nevertheless, prenatal administration of corticosterone impairs latent inhibition in males, but not in females [75]. Restraint PS could potentiate latent inhibition in the offspring [76], or have no effect [75, 77]. Exposure of pregnant rats to electric current results in disruption of latent inhibition in males, but not in females [75]. In our study, control males and females demonstrated a normal latent inhibition process, but US PS disrupted latent inhibition in males. This finding could potentially indicate some behavioral impairments since several studies have interpreted disrupted latent inhibition as an indicator decreased ability to use acquired patterns for learning [78].

US PS decreased the overall level of ASR in male rats, but not in females. Decreased ASR may indicate that the animal exhibits less fear, so the decreased ASR in the PS offspring can represent impaired behavioral reaction in response to the stress exposure [79]. Studies on animals have revealed that different types of PS mostly either have no effect on the strength of ASR, or increase its intensity [63, 80–82]. Nevertheless, the ASR reduction after the PS exposure has been shown in some studies. For example, juvenile male guinea pigs demonstrated significantly lower ASR after PS induced by a stroboscope compared to controls, but this was not observed in adult animals [79]. In another study, adult and adolescent rats showed a reduced ASR amplitude in the dexamethasone administration model [83].

Our study demonstrated that the US PS produced no effect on some behavioral traits. Rats after the US PS did not express depressive-like behavior. We also revealed no differences in the Morris water maze, indicating that the PS offspring demonstrated no impairments of spatial long-term memory. We investigated rats' short-term object memory in the object recognition test. PS males showed the decreased discrimination index, which may indicate memory impairment. However, this test showed a decrease of the total exploration time of the identical objects in the PS offspring. This fact cannot allow to evaluate these cognitive impairments in PS offspring definitively. Therefore, our study did not indicate a significant effect of US PS on rat's cognitive functions. We also demonstrated that US PS did not change PPI in rats. However, the results of this test may depend on the level of noise exposure during the test [84, 85]. In this connection, it is necessary to further investigate the US PS effect on the PPI with other parameters of the test protocol.

Changes of the Neurotransmitter Systems under the US PS Exposure

In our study, we determined biochemical parameters in the hippocampus and frontal cortex, which can reflect changes of the neurotransmitter systems functioning. We have demonstrated that US PS significantly affects their functioning. In addition, correlation analysis revealed the correlations between the behavior traits and neurotransmitter systems' activity. This supports the suggestion that neurotransmitter systems are involved in the development of pathological behavioral phenotype in the rat offspring after the US PS.

Exposure to US PS resulted in pronounced alterations of the serotonergic system in the hippocampus and frontal cortex. The concentration of serotonin increased in the frontal cortex, but its metabolic level did not alter. The

decrease of the HIAA/serotonin ratio, on the one hand, may indicate a decrease of serotonin metabolism. But since the HIAA concentration has not changed, this cannot be fair assumption. Most probably, the ratio change is related only to increased serotonin production and possibly to its decreased neurotransmission in frontal cortex. In contrast, PS offspring demonstrated elevated concentrations of both serotonin and HIAA in the hippocampus, which may indicate increased serotonin metabolism in this structure.

Elevated serotonin levels were associated predominantly with negative behavioral traits in animals from different groups, which probably indicate increased serotonin concentration as a pathological feature in PS offspring. Changes of the neurotransmitter systems functioning accompany various CNS pathologies. Numerous studies have demonstrated the involvement of the serotonergic system in the pathogenesis of mental disorders. Altered serotonin homeostasis at an early age is associated with various psychiatric pathologies, such as anxiety disorders, substance abuse, and especially autism [29, 30, 86]. Changes of the brain serotonergic system are also associated with depression. The dominant paradigm defines depression as a pathology that is accompanied by decreased serotonin levels in the brain. However, nowadays, a lot of data evidence to the contrary results [86], so this issue requires more research. Presumably, serotonin transmission may be increased in atypical forms of depression [87].

The serotonergic system has been studied previously in some animal PS models, which demonstrated changes of serotonergic system activity in some brain regions, but the results have been contradictory. PS induced opposite responses of the serotonergic system, depended on such factors as brain region, sex, age, and animal species and strain [88–90]. For example, it has been demonstrated that restraint PS can reduce serotonin metabolism in the cortex of adult mice [71] and in the hippocampus of juvenile mice [90], and decrease serotonin level in the brain of adult male Lewis rats, female Wistar and Lewis rats, and in female C57BL6 mice [88]. At the same time, administration of dexamethasone to pregnant females increased serotonin level in the medial prefrontal cortex of the adult offspring [91]. Nevertheless, another research of this model revealed decreased serotonin metabolism in the neocortex, hippocampus, hypothalamus, and mid-brain in juvenile and adult rats [89].

Therefore, in spite of the majority of studies show a decrease of serotonin level in different brain areas under the influence of PS, there is also the evidence of its increase, consistent with our research. Since it is known that serotonin is involved in many behavior reactions and also plays a role in neurodevelopment during the prenatal

period [17, 92], we can suggest that its increase may be involved in the processes by which US PS changes animal behavior in the postnatal period.

The US PS produced no significant effect on the dopaminergic system in the frontal cortex. The most pronounced differences were detected in the hippocampus. US PS increased dopamine concentration and decreased the DOPAC/dopamine ratio in the rat hippocampus, while the DOPAC concentration did not change, probably indicating the absence of dopamine metabolism changes in the hippocampus after US SP.

A few animal studies have demonstrated the effect of different PS types on the dopaminergic system in different areas of the brain. For example, administration of dexamethasone to pregnant female rats resulted in a significant decrease of dopamine metabolism in the hypothalamus [89]. In another study, restraint PS stimulated extracellular dopamine production in the nucleus accumbens of juvenile and adult rats [34]. Unpredictable chronic mild PS increased dopamine metabolism in the right prefrontal cortex and decreased dopamine activity in the right nucleus accumbens and left striatum of adult rats [93]. PS induced by tail suspension reduced dopamine metabolism in the nucleus accumbens of adult rat offspring [94]. Therefore, PS can potentially modulate the dopaminergic system of different brain regions, but the patterns of its effect can probably depend on various factors. Our study demonstrated the vulnerability of the hippocampal dopaminergic system to US PS, but the frontal cortex appeared to be less affected.

Altered catecholamine transmission plays an important role in the etiology of psychiatric disorders [34]. In particular, the dopamine hypothesis is the longest-standing pathoetiological theory of schizophrenia [28, 35]. Plenty of evidences of the dopamine system hyperactivity in schizophrenia exist. Antipsychotic drugs block dopamine receptors. Moreover, drugs that stimulate dopamine release or enhance its transmission, such as amphetamine or L-DOPA, exacerbate psychosis in patients with schizophrenia and can induce schizophrenic-like symptoms in healthy people [28, 95]. Postmortem studies have provided direct evidence of dopaminergic dysfunction in schizophrenia, showing elevated levels of dopamine, its metabolites, and receptors in the striatum [28]. Studies have also demonstrated changes of dopamine level and metabolic activity in certain brain areas (hippocampus, frontal cortex, striatum, hypothalamus, and substantia nigra) in animal models of schizophrenia-like state, obtained via the neurodevelopment impairments [26].

Interestingly, our experiment detected the increase of dopamine or its metabolite in the hippocampus of the PS offspring, which may indicate the presence of neuropathological features in rats after the US PS exposure comparable to the schizophrenia traits. It should also be mentioned that literature data evidence of the involvement of the hippocampus in the schizophrenia pathology. Human and animal studies have demonstrated that schizophrenia is characterized by decreased hippocampal volume and anterior hippocampal hyperactivity, which correlates with the severity of psychosis [95].

Finally, we demonstrated that US PS exposure provoked noradrenergic system changes in the rat offspring. PS offspring showed increased concentrations of noradrenaline in the frontal cortex and hippocampus. Some studies have revealed that different PS types can affect the noradrenergic system. For example, administration of dexamethasone to pregnant females resulted in decreased level of noradrenaline in the neocortex and hippocampus of the adult offspring [89]. PS induced by the daily subcutaneous injections of saline to pregnant female rats changed the concentration of norepinephrine in the hypothalamus and cortex of the offspring in an age-dependent manner [96]. PS induced by current shocks reduced the norepinephrine levels, but increased its metabolism in the cortex and locus coeruleus of the adult rats [97].

Sex Differences

Our study indicated some behavioral sex differences after US exposure. The research revealed the difference of the ASR parameter in the control group: males showed a higher level of ASR compared to the females. This difference was compensated after PS exposure. The existence of sex differences in the level of ASR in healthy rats is also supported by some studies [77, 98, 99]. The results of our experiment may indicate some influence of sex factor on the stress response in rats, which may be modulated by PS. Males subjected to US PS were characterized by a more anxious phenotype than females in the open-field test. The analysis of social behavior showed a greater vulnerability of social activity in males compared to the females. In addition, males after PS definitely demonstrated a disturbance of latent inhibition, which was not observed in females. Finally, the object recognition test showed that cognitive abilities of males were probably more susceptible to the negative influence of PS than in females. Therefore, we can conclude that males are more vulnerable to stress induced by US during the prenatal period. These results are consistent with many

studies that have used other types of PS in animal experiments [100–103], suggesting that important sex differences in fetal responses to maternal stress during pregnancy exist.

Sex differences in biochemical parameters were also observed in our research. Multiple biochemical parameters significantly differed between males and females in the control animals (HIAA, HIAA/serotonin ratio, DOPAC, DOPAC/dopamine ratio, and α -MSH). Nevertheless, these differences were not detected in PS offspring. It was shown that PS can disturb sexual differentiation of the brain. Most frequently, these impairments can shift the brain developing toward demasculinization or feminization in males [104, 105]. The neurotransmitter systems of the brain play an extremely important role in this process. Neurotransmitters sexual differentiation largely determines the sexual dimorphism of the whole neuroendocrine system [89, 105, 106]. These assumptions are also supported by the results of our study as the effects of US PS alleviated the sex differences of neurochemical and behavioral parameters, associated with each other.

Therefore, our findings highlight the importance of including both sexes into scientific researches. It is also important to develop future therapeutic strategies for humans that consider gender. Data from animal studies represent a substantial step toward this goal.

Modeling Mental Disorders by the US PS

In general, the results of our study demonstrated the negative effect of US exposure during the prenatal period on the rat's behavior, which is consistent with the results of previous studies of other prenatal stressors. As rodents communicate via US, we suggest that this type of stressor can have psychological character. In view of this, the use of US PS can probably be applied to model PS with psychological nature. As such, stress is most typical for humans in modern society; our translational model has the potential to describe more appropriately the effect of stress during pregnancy compared to the models with other stress types. Such models are mostly based on the physical stress exposure, often used in experiments with laboratory animals.

PS is a risk factor for the development of mental disorders in humans, so PS is used to model mental disorders on animals. Some attempts were undertaken to develop PS-based models of diseases such as schizophrenia and depression [107, 108]. Since US PS leads to a behavioral dysfunction in animals and shows similarities with some symptoms of psychopathologies in humans,

potentially this type of stress can be used to model mental disorders.

The impaired behavior of animals after US PS exposure could indicate the external validity of our model. In particular, our study revealed that US PS increase anxious behavior, alter social activity, and exploratory behavior in animals. These behavioral dysfunctions may be associated with certain symptoms in humans, but they are not disease-specific and characterize a wide range of human mental disorders. For example, social dysfunction is a defining feature of autism [109], but impaired social interaction can also be observed in schizophrenia [110], social anxiety disorder [111], and depression [112]. Increased anxiety is characteristic for the anxiety personality disorder, but it can also accompany autism [113] and schizophrenia [114]. Disturbance of orientation and exploratory activity, namely, reactions to unexpected stimuli and novelty, can be observed during neurosis, organic brain damage, schizophrenia, and depression [115, 116].

The social impairments in the social interaction test can indicate the presence of autistic-like behavior in the animals after US PS. From this point of view, it is also important to consider the identified relationship between anxious and social behavior. In adult control males, social behavior was negatively correlated with the anxious behavior. As the social interaction test was developed as one of the first models of anxiety-like behavior and the reduction of social activity in this test indicated the increased anxiety [117], this relationship is predictable. It is suggested that a decrease of social interaction indicates social avoidance and thus may reflect social fear in rodents [118]. It is interesting that impaired social behavior in males after PS was not associated with the level of anxiety. However, their level of anxiety was correlated with the exploration time of nonliving object that probably can reflect the deviance in behavioral reactions from social interaction to interaction with nonliving objects, which in turn could possibly indicate the presence of quasi-autistic traits in rodent's behavior since patients with autism are characterized by the modification of attention from social toward non-social stimuli [119]. At the same time, our previous studies have demonstrated that US PS altered social preferences, decreased play behavior in juvenile rats [46], and decreased social contact in adult Sprague-Dawley rats [47]. Impaired social interaction is one of the major characteristics of autism [120]. Animal autism models are expected to exhibit at least one of the main behavioral symptoms: impaired social interaction or the presence of stereotyped repetitive behavior [121]. This

fact allow us to suggest the presence of autism-like behavior phenotype in rats after US PS, especially given the fact that social dysfunction was observed from the early age [46], when relevant autism symptoms are also observed in humans.

Another frequent symptom of autism is autoaggression [122]. Increased autoaggressive behavior and other types of behavior directed at their own body (grooming, scratching) can also be observed in animals [123]. Although grooming is not a defining trait of any modeling pathology, its various aspects can reflect some patterns of human behavior [124]. Some evidence suggests that increased grooming activity may characterize models of autism, anxiety disorder, psychosis, and obsessive-compulsive disorder [68, 124–127].

In order to investigate depressive-like behavior in the animals in our experiment, we conducted behavioral tests that could detect anhedonia and despair behavior in animals since these behavioral traits are commonly associated with symptoms of depression in humans [108]. However, these tests showed the absence of behavioral abnormalities, which allowed us to exclude a depressive-like behavioral phenotype in animals after US PS. In addition, the study of the serotonergic system has not demonstrated a decrease of serotonin level in the frontal cortex or hippocampus characteristic for depression. Instead, the activation of this system was revealed after the US PS. However, in our study, we detected a decrease of ASR in males exposed to US PS. This phenomenon can also be observed in people suffering from mental disorders. Patients with depression or bipolar disorder demonstrated a blunted response to acoustic stimuli [128, 129]. Patients with distress states also showed signs of reduced ASR [130, 131]. Therefore, depression may produce an inhibitory effect on the startle modulation. This allows to suggest the presence of some depressive-like behavioral traits in animals after US PS.

The important result was shown by the latent inhibition test, which detected impaired latent inhibition process in rats after US PS. This trait is often observed in patients with schizophrenia and is normalized by neuroleptics [76]. It is suggested that latent inhibition may be associated with positive symptoms of schizophrenia [26]. Latent inhibition has been widely studied in translational schizophrenia models and is closely related to dopaminergic activity [76, 132]. Most rodent models of schizophrenia primarily reproduce positive symptoms of schizophrenia as negative and cognitive symptoms are less specific [107]. Nevertheless, schizophrenia represents a complex disease with a heterogeneous group of symptoms, which cannot be fully modeled in rodents

[132, 133]. Model validity is stronger when behavioral phenotypes reflect two or more disease symptoms [133]. In our study, beside the observed latent inhibition, which can be compared to positive symptomatology, social dysfunction was also detected in animals after US PS, which may be associated with negative symptoms. We observed no pronounced cognitive impairment in rats, nonetheless schizophrenia is not always characterized by cognitive decline [134, 135]. Thus, it is possible to suggest that impaired behavioral phenotype in animals after US PS may reflect schizophrenic-like behavior features. The additional important confirmation of this assumption is the detected activation of the dopaminergic system in the rat hippocampus since this phenomenon is characteristic for schizophrenia.

It is important to note that the previous suggestion about autistic-like behavior observed in rats in our experiment does not contradict this assumption since the psychopathology of schizophrenia and autism significantly overlap [26]. Because of the similarities in social deficits that can be observed in rodent models of schizophrenia and autism, it is difficult to distinguish between the social dysfunction associated with the two disorders [136].

Nevertheless, the potential model of a schizophrenic-like condition has the advantage because of the negative symptoms presence. While current therapeutic agents (neuroleptics) can reduce the positive symptoms of schizophrenia, treatment of the negative symptoms is a vital and unmet clinical need [107]. Developing of the translational models that more appropriately replicate the negative disease symptoms, in addition to positive symptoms, is an important task [107].

Additionally, it is necessary to mention schizotypal disorder. Schizotypal personality traits phenomenologically remind the signs and symptoms of schizophrenia and conceptualized as a latent predisposition to schizophrenia [137]. Schizotypal disorder is generally characterized by three major types of symptoms: cognitive deficits of perception (strange beliefs, perceptual disturbances, and mentalism), interpersonal deficits (lack of intimate friends, social anxiety), and disorganization (strange speech and thoughts, blunted affect, odd behavior) [138, 139]. This set of personality traits can range from minimal disturbances to psychotic disorders in the general human population [140]. A number of studies have shown that latent inhibition can also be impaired in this disorder [141, 142], which allows us to consider the behavioral phenotype of rats after US PS also as a model, reproducing schizotypy symptoms, that may have important practical relevance as schizotypy allows us to

explore the etiological pathways underlying schizophrenic spectrum psychopathology [139].

The promising experimental approach to focus on individual behavioral, physiological, and neuroanatomical phenotypes of the disease, rather than modeling the entire syndrome, should also be mentioned [26]. For the great majority of psychiatric pathological conditions, knowledge of pathophysiology remains insufficient, and objective diagnostic tests for them are lacking. Diagnosis in psychiatry is currently guided by the International Classification of Diseases (ICD) and the Diagnostic and Statistical Manual of Mental Disorders (DSM), which are based on the assumption that mental disorders represent separated nosological entities. Diagnoses are based exclusively on phenomenology, namely, on symptoms, signs, and progress of the disease. However, the similarity of clinical phenotypic manifestations and frequently observed comorbidities support the assumption that mental disorders are not separate conditions [26, 135, 143]. DSM and ICD do not correlate well with the new data from genetics and neurobiology, hindering not only etiology and pathophysiology research but also the development of the new treatments [144]. Psychiatric drugs are not effective in all patients. One reason for this low response rate is the artificial association of heterogeneous syndromes with different pathophysiological mechanisms into a common disorder [144].

Because of this, in 2009, the National Institute of Mental Health launched the Research Domain Criteria (RDoC) project [134, 144] to develop a system for classifying mental disorders based on neurobiology parameters and observed behavior, which represent a new approach to translational research. The RDoC represents the paradigm allowing to look at mental disorders from a translational perspective. RDoC enhances the translational potential of animal research to address neurobiological questions related to psychopathology. The RDoC structure avoids the serious problem of translational research strictly conforming to an animal model of disorder [134].

The borders between the different disorders, as well as the boundaries between the pathology and mental normality, are often arbitrary, creating huge barriers to the development of effective animal models of mental disorders. Animal models cannot fully reflect human mental disorder characteristics. In addition, individual symptoms observed in animal models may not have a simple and direct correspondence to the human symptoms. Translational models reproduce only certain symptoms instead of accurate and complete simulations of psychiatric syndromes [26, 132, 135]. Therefore, we suggest

that it is not necessary to associate behavioral phenotype impairments in rats after US PS with a strictly defined psychopathological syndrome. Instead, we need to look more closely at individual symptoms, such as social deficits, increased anxious behavior, or impaired latent Inhibition, as well as changes of neurotransmitter systems. Such complex may form the core or be one of the elements of pathological manifestations of different diseases and represent the target for therapy of different nosologies.

Conclusion

Our study demonstrated that stress in pregnant females caused by chronic exposure to variable frequency US, which carries the information significance for rodents, lead to the development of the pathological behavioral phenotype in the offspring. Emotional reactions, social functions, and exploratory behavior of the animals were mostly affected. These behavioral disturbances were accompanied by changes in the functioning of the serotonergic, dopaminergic, and noradrenergic neurotransmitter systems in the frontal cortex and hippocampus, which were correlated with behavioral dysfunction.

Our study has practical relevance because PS in rodents caused by variable US exposure could probably be applied to model the psychological PS that is primary characteristic for humans in modern society. In addition, US PS leads to the certain dysfunction on behavioral and biochemical levels in animals, which could potentially indicate the validity for future modeling of psychopathology based on the impairments of neurodevelopment. Our study has shown that changes in rats after US PS are most closely associated with symptoms of schizophrenia/schizotypy or autism. However, in order to develop a complete effective translational model, predictive, and construct validity must also be evaluated in the future.

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Statement of Ethics

This study protocol was reviewed and approved by Local Ethical Committee of V.P. Serbsky National Medical Research Center for Psychiatry and Narcology, approval number 1 from

February 10, 2022. Housing conditions and all experimental procedures were set up, maintained, conducted in accordance with Directive 2010/63/EU of September 22, 2010.

Conflict of Interest Statement

The authors have no conflicts of interest to declare.

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Author Contributions

Conceptualization: E.Z., A.M., and V.C.; methodology: E.Z. and A.M.; formal analysis: O.A. and Y.Z.; investigation: O.A., Y.Z., V.U., and A.T.P.; resources: O.G. and V.C.; writing – original draft preparation: O.A., Y.Z., and V.U.; writing – review and editing, O.A., Y.Z., E.Z., Z.S., A.M., and V.U.; supervision: O.G., Z.S., and V.C.; project administration: E.Z. and A.M.; funding acquisition: O.G.

Data Availability Statement

All data generated or analyzed during this study are included in this article. Further inquiries can be directed to the corresponding author.

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